

Numerical Analysis homework # 6

Guang Zhang 3230105121 *

(Information and Computing Science, Class XJ2301), Zhejiang University

Finish time: December 25, 2025

I. Question I

I.a

We seek to approximate the integral $I = \int_{-1}^1 y(t)dt$ using a cubic Hermite interpolating polynomial $p_3(t)$ satisfying the conditions:

$$p_3(-1) = y(-1), \quad p_3(0) = y(0), \quad p_3'(0) = y'(0), \quad p_3(1) = y(1).$$

Let the polynomial be explicitly defined as $p_3(t) = a_0 + a_1t + a_2t^2 + a_3t^3$. Integrating this polynomial over $[-1, 1]$:

$$\int_{-1}^1 p_3(t)dt = \int_{-1}^1 (a_0 + a_1t + a_2t^2 + a_3t^3)dt.$$

Due to the symmetry of the interval $[-1, 1]$, the integrals of the odd power terms (t and t^3) vanish. We are left with:

$$\int_{-1}^1 p_3(t)dt = \left[a_0t + \frac{1}{3}a_2t^3 \right]_{-1}^1 = 2a_0 + \frac{2}{3}a_2.$$

Now we determine the coefficients a_0 and a_2 using the interpolation conditions:

1. At $t = 0$: $p_3(0) = a_0 = y(0)$.
2. At $t = 1$ and $t = -1$:

$$\begin{cases} p_3(1) = a_0 + a_1 + a_2 + a_3 = y(1) \\ p_3(-1) = a_0 - a_1 + a_2 - a_3 = y(-1) \end{cases}$$

Adding these two equations eliminates a_1 and a_3 :

$$y(1) + y(-1) = 2a_0 + 2a_2.$$

Substituting $a_0 = y(0)$, we solve for a_2 :

$$2a_2 = y(1) + y(-1) - 2y(0) \implies a_2 = \frac{1}{2}[y(1) + y(-1)] - y(0).$$

Substituting a_0 and a_2 back into the integral expression:

$$\begin{aligned} \int_{-1}^1 p_3(t)dt &= 2y(0) + \frac{2}{3} \left(\frac{1}{2}[y(1) + y(-1)] - y(0) \right) \\ &= 2y(0) + \frac{1}{3}[y(1) + y(-1)] - \frac{2}{3}y(0) \\ &= \frac{4}{3}y(0) + \frac{1}{3}[y(1) + y(-1)] \\ &= \frac{1}{3}[y(-1) + 4y(0) + y(1)]. \end{aligned}$$

*Electronic address: zhuiguang49@foxmail.com

This is exactly the formulation of Simpson's rule. Actually, we can obtain Simpson's rule by Lagrange base function, we have

$$l_0(x) = \frac{x(x-1)}{2} \quad (1)$$

$$l_1(x) = \frac{-2}{x} + 1 \quad (2)$$

$$l_2(x) = \frac{x(x+1)}{2} \quad (3)$$

From which we can obtain that $A_0 = \frac{1}{3}, A_1 = \frac{4}{3}, A_2 = \frac{1}{3}$, which leads to Simpson's rule:

$$\int_{-1}^1 y(t)dt = \frac{1}{3}[y(-1) + 4y(0) + y(1)] \quad (4)$$

I.b

The error of the Hermite interpolation $p_3(t)$ for a function $y \in C^4[-1, 1]$ is given by:

$$y(t) - p_3(t) = \frac{y^{(4)}(\xi_t)}{4!}\Omega(t),$$

where the node polynomial is $\Omega(t) = (t - (-1))(t - 0)^2(t - 1) = (t^2 - 1)t^2$. The error of the quadrature is obtained by integrating this remainder:

$$E^S(y) = \int_{-1}^1 [y(t) - p_3(t)]dt = \int_{-1}^1 \frac{y^{(4)}(\xi_t)}{24} t^2(t^2 - 1)dt.$$

We observe that on the interval $[-1, 1]$, the term $t^2 \geq 0$ and $t^2 - 1 \leq 0$. Thus, the kernel function $K(t) = t^2(t^2 - 1)$ does not change sign (it is non-positive). By the Mean Value Theorem for Integrals, there exists some $\zeta \in (-1, 1)$ such that:

$$E^S(y) = \frac{y^{(4)}(\zeta)}{24} \int_{-1}^1 (t^4 - t^2)dt.$$

Computing the definite integral:

$$\int_{-1}^1 (t^4 - t^2)dt = \left[\frac{t^5}{5} - \frac{t^3}{3} \right]_{-1}^1 = 2 \left(\frac{1}{5} - \frac{1}{3} \right) = 2 \left(\frac{3-5}{15} \right) = -\frac{4}{15}.$$

Therefore, the error is:

$$E^S(y) = \frac{y^{(4)}(\zeta)}{24} \left(-\frac{4}{15} \right) = -\frac{1}{90} y^{(4)}(\zeta).$$

I.c

To derive the composite Simpson's rule for $\int_a^b f(x)dx$, we partition $[a, b]$ into n subintervals (where n is even) with step size $h = \frac{b-a}{n}$ and nodes $x_k = a + kh$. We consider the sub-interval $[x_{2k}, x_{2k+2}]$ of width $2h$.

We perform a change of variables $x = x_{2k+1} + th$, where $t \in [-1, 1]$. Then $dx = hdt$. Let $y(t) = f(x_{2k+1} + th)$. The integral over one panel is:

$$I_k = \int_{x_{2k}}^{x_{2k+2}} f(x)dx = h \int_{-1}^1 y(t)dt.$$

Using the result from (a), the approximation is:

$$I_k \approx h \cdot \frac{1}{3}[y(-1) + 4y(0) + y(1)] = \frac{h}{3}[f(x_{2k}) + 4f(x_{2k+1}) + f(x_{2k+2})].$$

Summing over all $k = 0, \dots, n/2 - 1$, we get the composite formula:

$$I_n^S(f) = \sum_{k=0}^{n/2-1} I_k = \frac{h}{3} \left[f(a) + 4 \sum_{k=0}^{n/2-1} f(x_{2k+1}) + 2 \sum_{k=1}^{n/2-1} f(x_{2k}) + f(b) \right].$$

Error Estimate: From part (b), the error on the standard interval is $E^S(y) = -\frac{1}{90}y^{(4)}(\zeta_k)$. By the Chain Rule, derivatives transform as $y^{(4)}(t) = h^4 f^{(4)}(x)$. Thus, the error for the k -th sub-interval is:

$$E_k = h \cdot E^S(y) = h \left(-\frac{1}{90} h^4 f^{(4)}(\xi_k) \right) = -\frac{h^5}{90} f^{(4)}(\xi_k), \quad \xi_k \in (x_{2k}, x_{2k+2}).$$

The total error is the sum of errors over the $n/2$ sub-intervals:

$$E_{total} = \sum_{k=0}^{n/2-1} E_k = -\frac{h^5}{90} \sum_{k=0}^{n/2-1} f^{(4)}(\xi_k).$$

Using the Intermediate Value Theorem, since $f^{(4)}$ is continuous, $\sum_{k=0}^{n/2-1} f^{(4)}(\xi_k) = \frac{n}{2} f^{(4)}(\xi)$ for some $\xi \in (a, b)$. Substituting $n = \frac{b-a}{h}$:

$$E_{total} = -\frac{h^5}{90} \cdot \frac{n}{2} f^{(4)}(\xi) = -\frac{h^5}{90} \cdot \frac{b-a}{2h} f^{(4)}(\xi) = -\frac{b-a}{180} h^4 f^{(4)}(\xi).$$

II. Question II

We aim to approximate the integral $I = \int_0^1 e^{-x^2} dx$ with an absolute error strictly less than $\epsilon = 0.5 \times 10^{-6}$. Let $f(x) = e^{-x^2}$. The interval is $[a, b] = [0, 1]$.

II.a

The error bound for the composite trapezoidal rule is given by:

$$|E_n^T| \leq \frac{(b-a)^3}{12n^2} \max_{x \in [a,b]} |f''(x)|.$$

First, we compute the second derivative of $f(x)$:

$$\begin{aligned} f'(x) &= -2xe^{-x^2} \\ f''(x) &= (4x^2 - 2)e^{-x^2} \end{aligned}$$

By calculation $M_2 = \max_{x \in [0,1]} |f''(x)| = 2$. Substituting into the error inequality:

$$\begin{aligned} \frac{(1-0)^3}{12n^2} \cdot 2 &< 0.5 \times 10^{-6} \\ \frac{1}{6n^2} &< 5 \times 10^{-7} \\ n^2 &> \frac{1}{30 \times 10^{-7}} \approx 333,333.33 \\ n &> \sqrt{333,333.33} \approx 577.35 \end{aligned}$$

Since n must be an integer, the required number of subintervals is:

$$n = 578.$$

II.b

The error bound for the composite Simpson's rule is given by:

$$|E_n^S| \leq \frac{(b-a)^5}{180n^4} \max_{x \in [a,b]} |f^{(4)}(x)|.$$

We determine the fourth derivative $f^{(4)}(x)$:

$$\begin{aligned} f'''(x) &= (8x - 8x^3 + 4x)e^{-x^2} = (12x - 8x^3)e^{-x^2} \\ f^{(4)}(x) &= 4(4x^4 - 12x^2 + 3)e^{-x^2} \end{aligned}$$

By caculation $M_4 = \max_{x \in [0,1]} |f^{(4)}(x)| = 12$. Substituting into the error inequality:

$$\begin{aligned}\frac{(1-0)^5}{180n^4} \cdot 12 &< 0.5 \times 10^{-6} \\ \frac{1}{15n^4} &< 5 \times 10^{-7} \\ n^4 &> \frac{1}{75 \times 10^{-7}} \approx 133,333.33 \\ n &> \sqrt[4]{133,333.33} \approx 19.1\end{aligned}$$

For Simpson's rule, n must be an even integer. The smallest even integer greater than 19.1 is:

$$n = 20.$$

III. Question III

III.a

We seek a polynomial $\pi_2(t) = t^2 + at + b$ that is orthogonal to P_1 with respect to the weight function $\rho(t) = e^{-t}$ on $[0, +\infty)$. The orthogonality conditions are:

$$\langle \pi_2, 1 \rangle = 0 \quad \text{and} \quad \langle \pi_2, t \rangle = 0.$$

Using the property $\int_0^{+\infty} t^m e^{-t} dt = m!$, we compute the inner products:

$$\int_0^{+\infty} (t^2 + at + b)e^{-t} dt = 2! + a(1!) + b(0!) = 2 + a + b = 0 \quad (5)$$

$$\int_0^{+\infty} t(t^2 + at + b)e^{-t} dt = \int_0^{+\infty} (t^3 + at^2 + bt)e^{-t} dt = 3! + 2!a + 1!b = 6 + 2a + b = 0 \quad (6)$$

Then we solve the linear system which leads to $a = -4, b = 2$. Thus, the orthogonal polynomial is:

$$\pi_2(t) = t^2 - 4t + 2.$$

III.b

1. Finding the Nodes: The nodes t_1, t_2 are the roots of $\pi_2(t) = 0$:

$$t = \frac{4 \pm \sqrt{16 - 8}}{2} = 2 \pm \sqrt{2}.$$

Let $t_1 = 2 - \sqrt{2}$ and $t_2 = 2 + \sqrt{2}$.

2. Finding the Weights: The quadrature formula is exact for degree $2n - 1 = 3$. We can determine the weights w_1, w_2 by enforcing exactness for $f(t) = 1$ and $f(t) = t$:

$$\begin{cases} w_1 + w_2 = \int_0^{+\infty} e^{-t} dt = 1 \\ w_1 t_1 + w_2 t_2 = \int_0^{+\infty} t e^{-t} dt = 1 \end{cases}$$

Solving this system:

$$w_1(2 - \sqrt{2}) + w_2(2 + \sqrt{2}) = 1 \implies 2(w_1 + w_2) + \sqrt{2}(w_2 - w_1) = 1.$$

Since $w_1 + w_2 = 1$, we have $2 + \sqrt{2}(w_2 - w_1) = 1 \implies w_2 - w_1 = -\frac{1}{\sqrt{2}}$. Combined with $w_1 + w_2 = 1$, we get:

$$w_1 = \frac{2 + \sqrt{2}}{4}, \quad w_2 = \frac{2 - \sqrt{2}}{4}.$$

The formula is:

$$\int_0^{+\infty} f(t)e^{-t} dt \approx \frac{2 + \sqrt{2}}{4} f(2 - \sqrt{2}) + \frac{2 - \sqrt{2}}{4} f(2 + \sqrt{2}).$$

3. Error Term: The remainder for Gaussian quadrature is given by:

$$E_2(f) = \frac{f^{(4)}(\tau)}{(2n)!} \int_0^{+\infty} [\pi_2(t)]^2 e^{-t} dt.$$

First, compute the integral constant:

$$\int_0^{+\infty} (t^2 - 4t + 2)^2 e^{-t} dt = \int_0^{+\infty} (t^4 - 8t^3 + 20t^2 - 16t + 4) e^{-t} dt.$$

Using $\int t^m e^{-t} = m!$:

$$= 24 - 8(6) + 20(2) - 16(1) + 4 = 24 - 48 + 40 - 16 + 4 = 4.$$

Thus, the error term is:

$$E_2(f) = \frac{f^{(4)}(\tau)}{24} \cdot 4 = \frac{1}{6} f^{(4)}(\tau), \quad \tau > 0.$$

III.c

We approximate $I = \int_0^{+\infty} \frac{1}{1+t} e^{-t} dt$. Here $f(t) = \frac{1}{1+t}$. **1. Approximation:** Using the formula derived in (b):

$$\begin{aligned} I_{\text{approximate}} &= w_1 f(t_1) + w_2 f(t_2) \\ &= \frac{2 + \sqrt{2}}{4} \cdot \frac{1}{1 + (2 - \sqrt{2})} + \frac{2 - \sqrt{2}}{4} \cdot \frac{1}{1 + (2 + \sqrt{2})} \\ &= \frac{2 + \sqrt{2}}{4(3 - \sqrt{2})} + \frac{2 - \sqrt{2}}{4(3 + \sqrt{2})} \end{aligned}$$

Simplifying the terms:

$$\begin{aligned} \frac{2 + \sqrt{2}}{4(3 - \sqrt{2})} \cdot \frac{3 + \sqrt{2}}{3 + \sqrt{2}} &= \frac{6 + 5\sqrt{2} + 2}{4(9 - 2)} = \frac{8 + 5\sqrt{2}}{28} \\ \frac{2 - \sqrt{2}}{4(3 + \sqrt{2})} \cdot \frac{3 - \sqrt{2}}{3 - \sqrt{2}} &= \frac{6 - 5\sqrt{2} + 2}{4(9 - 2)} = \frac{8 - 5\sqrt{2}}{28} \end{aligned}$$

Summing them up:

$$I_{\text{approx}} = \frac{16}{28} = \frac{4}{7} \approx 0.57142857.$$

2. Comparison and Identification of τ : Given the exact value $I_{\text{true}} = 0.59634736$, the true error is:

$$E = I_{\text{true}} - I_{\text{approx}} = 0.59634736 - 0.57142857 = 0.02491879.$$

From (b), the theoretical error is $E_2(f) = \frac{1}{6} f^{(4)}(\tau)$. We compute derivatives of $f(t) = (1 + t)^{-1}$:

$$f'(t) = -(1 + t)^{-2}, \quad \dots, \quad f^{(4)}(t) = 24(1 + t)^{-5}.$$

Thus, $E_2(f) = \frac{1}{6} \cdot 24(1 + \tau)^{-5} = 4(1 + \tau)^{-5}$. Equating the true error to the theoretical form:

$$4(1 + \tau)^{-5} = 0.02491879 \implies (1 + \tau)^5 = \frac{4}{0.02491879} \approx 160.521.$$

Solving for τ :

$$1 + \tau = (160.521)^{1/5} \approx 2.766 \implies \tau \approx 1.76125$$

This value $\tau \approx 1.76125$ lies within the integration domain $(0, +\infty)$, confirming the validity of the error term.

IV. Question IV

IV.a

We seek basis functions of the form:

$$h_m(t) = (a_m + b_m t) l_m^2(t), \quad q_m(t) = (c_m + d_m t) l_m^2(t).$$

Recall the properties of the Lagrange basis $l_m(t)$:

$$l_m(x_j) = \delta_{mj}.$$

Differentiation yields $h'_m(t) = b_m l_m^2(t) + (a_m + b_m t) 2l_m(t) l'_m(t)$. **1. Determining $h_m(t)$:** We require $h_m(x_m) = 1$ and $h'_m(x_m) = 0$.

- $h_m(x_m) = (a_m + b_m x_m) \cdot 1^2 = 1 \implies a_m + b_m x_m = 1.$
- $h'_m(x_m) = b_m \cdot 1 + (a_m + b_m x_m) \cdot 2l'_m(x_m) = 0.$

Substituting $a_m + b_m x_m = 1$ into the derivative equation:

$$b_m + 2(1)l'_m(x_m) = 0 \implies b_m = -2l'_m(x_m).$$

Then $a_m = 1 - b_m x_m = 1 + 2x_m l'_m(x_m)$. Thus, the coefficients are $a_m = 1 + 2x_m l'_m(x_m)$ and $b_m = -2l'_m(x_m)$. Alternatively written:

$$h_m(t) = [1 - 2(t - x_m)l'_m(x_m)]l_m^2(t).$$

2. Determining $q_m(t)$: We require $q_m(x_m) = 0$ and $q'_m(x_m) = 1$.

- $q_m(x_m) = (c_m + d_m x_m) \cdot 1 = 0 \implies c_m = -d_m x_m.$
- $q'_m(x_m) = d_m + 2(c_m + d_m x_m)l'_m(x_m) = 1.$

Since $c_m + d_m x_m = 0$, the second term vanishes:

$$d_m + 0 = 1 \implies d_m = 1.$$

Then $c_m = -x_m$. Thus, the coefficients are $c_m = -x_m$ and $d_m = 1$. This gives:

$$q_m(t) = (t - x_m)l_m^2(t).$$

IV.b

Let $p(t)$ be the Hermite interpolating polynomial for $f(t)$. Integrating $p(t)$ gives the quadrature rule:

$$I_n(f) = \int_a^b p(t)\rho(t)dt = \sum_{k=1}^n \left[f(x_k) \int_a^b h_k(t)\rho(t)dt + f'(x_k) \int_a^b q_k(t)\rho(t)dt \right].$$

Defining the weights and coefficients as:

$$w_k = \int_a^b h_k(t)\rho(t)dt, \quad \mu_k = \int_a^b q_k(t)\rho(t)dt.$$

We obtain the rule:

$$I_n(f) = \sum_{k=1}^n [w_k f(x_k) + \mu_k f'(x_k)].$$

Since $p(t)$ reproduces any polynomial in P_{2n-1} exactly, the error $E_n(p) = 0$ for all $p \in P_{2n-1}$.

IV.c

To eliminate the derivative terms (i.e., $\mu_k = 0$ for all k), we examine the expression for μ_k :

$$\mu_k = \int_a^b (t - x_k)l_k^2(t)\rho(t)dt.$$

Let $v_n(t) = \prod_{j=1}^n (t - x_j)$ be the node polynomial. Recall that the Lagrange basis can be written as $l_k(t) = \frac{v_n(t)}{(t - x_k)v'_n(x_k)}$. Substituting one factor of $l_k(t)$ into the integral:

$$(t - x_k)l_k^2(t) = (t - x_k) \frac{v_n(t)}{(t - x_k)v'_n(x_k)} l_k(t) = \frac{1}{v'_n(x_k)} v_n(t) l_k(t).$$

Thus,

$$\mu_k = \frac{1}{v'_n(x_k)} \int_a^b v_n(t) l_k(t) \rho(t) dt.$$

For μ_k to vanish for all k , the integral $\int_a^b v_n(t) l_k(t) \rho(t) dt$ must be zero. Since $\{l_k(t)\}_{k=1}^n$ forms a basis for P_{n-1} , the condition $\int v_n l_k \rho dt = 0$ for all k is equivalent to saying that $v_n(t)$ is orthogonal to the entire space P_{n-1} with respect to the weight $\rho(t)$.

Conclusion: The condition required is that the node polynomial $v_n(t)$ must be the **orthogonal polynomial** of degree n with respect to the weight function $\rho(t)$. This implies that the nodes x_k must be the roots of this orthogonal polynomial (i.e., they must be the **Gauss nodes**). Under this condition, the quadrature formula becomes the standard Gaussian Quadrature with degree of exactness $2n - 1$.

V. Question V

V.1 Proof of Lemma 6.40

We consider the second-order central difference formula for the second derivative:

$$D^2u(\bar{x}) = \frac{u(\bar{x} - h) - 2u(\bar{x}) + u(\bar{x} + h)}{h^2}.$$

1. Truncation Error: Assuming $u \in C^4$, we expand $u(\bar{x} \pm h)$ using Taylor series about $\bar{x}$:

$$u(\bar{x} \pm h) = u(\bar{x}) \pm hu'(\bar{x}) + \frac{h^2}{2}u''(\bar{x}) \pm \frac{h^3}{6}u'''(\bar{x}) + \frac{h^4}{24}u^{(4)}(\xi_{\pm}).$$

Summing the two expansions:

$$u(\bar{x} + h) + u(\bar{x} - h) = 2u(\bar{x}) + h^2u''(\bar{x}) + \frac{h^4}{24}(u^{(4)}(\xi_+) + u^{(4)}(\xi_-)).$$

By the Intermediate Value Theorem, $\frac{1}{2}(u^{(4)}(\xi_+) + u^{(4)}(\xi_-)) = u^{(4)}(\xi)$ for some $\xi \in (\bar{x} - h, \bar{x} + h)$. Thus:

$$u(\bar{x} + h) - 2u(\bar{x}) + u(\bar{x} - h) = h^2u''(\bar{x}) + \frac{h^4}{12}u^{(4)}(\xi).$$

Dividing by h^2 gives the truncation error:

$$D^2u(\bar{x}) - u''(\bar{x}) = \frac{h^2}{12}u^{(4)}(\xi).$$

2. Rounding Error: Let $\tilde{u}$ be the computed values with rounding error $|\tilde{u} - u| \leq \epsilon$. The computed difference is:

$$D^2\tilde{u} = \frac{\tilde{u}(\bar{x} - h) - 2\tilde{u}(\bar{x}) + \tilde{u}(\bar{x} + h)}{h^2}.$$

The error due to rounding is bounded by:

$$|E_{\text{round}}| \leq \frac{|\epsilon| + |-2\epsilon| + |\epsilon|}{h^2} = \frac{4\epsilon}{h^2}.$$

3. Total Error Bound: Combining both errors, the total error bound is:

$$|E(h)| \leq \frac{h^2}{12}M_4 + \frac{4\epsilon}{h^2}, \quad \text{where } M_4 = \max |u^{(4)}(x)|.$$

4. Minimizing the Error: To find the optimal h , we define $\phi(h) = \frac{M_4}{12}h^2 + \frac{4\epsilon}{h^2}$ and set $\phi'(h) = 0$:

$$\frac{M_4h}{6} - \frac{8\epsilon}{h^3} = 0 \implies h^4 = \frac{48\epsilon}{M_4} \implies h_{\text{opt}} = \left(\frac{48\epsilon}{M_4}\right)^{1/4}.$$

The minimal error scales as $O(\epsilon^{1/2})$.

V.2 Designing a Fourth-Order Symmetric Formula

We construct a formula using a 5-point symmetric stencil: $\bar{x}, \bar{x} \pm h, \bar{x} \pm 2h$.

$$D_4^2u(\bar{x}) = \frac{c_0u(\bar{x}) + c_1[u(\bar{x} + h) + u(\bar{x} - h)] + c_2[u(\bar{x} + 2h) + u(\bar{x} - 2h)]}{h^2}.$$

Expanding the terms using Taylor series:

$$\begin{aligned} u(\bar{x} \pm h) + u(\bar{x} \mp h) &= 2u + h^2u'' + \frac{h^4}{12}u^{(4)} + \frac{h^6}{360}u^{(6)} + \dots \\ u(\bar{x} \pm 2h) + u(\bar{x} \mp 2h) &= 2u + 4h^2u'' + \frac{16h^4}{12}u^{(4)} + \frac{64h^6}{360}u^{(6)} + \dots \end{aligned}$$

Substitute into the formula and match coefficients to approximate $1 \cdot u''$:

$$u : c_0 + 2c_1 + 2c_2 = 0 \quad (7)$$

$$u'' : c_1 + 4c_2 = 1 \quad (8)$$

$$u^{(4)} : \frac{1}{12}c_1 + \frac{16}{12}c_2 = 0 \implies c_1 + 16c_2 = 0 \quad (9)$$

We can finally obtain that the formula is:

$$D_4^2 u(\bar{x}) = \frac{-\frac{1}{12}u_{\pm 2} + \frac{16}{12}u_{\pm 1} - \frac{30}{12}u_0}{h^2} = \frac{-u_{\pm 2} + 16u_{\pm 1} - 30u_0}{12h^2}.$$

1. Truncation Error Bound: The leading error term comes from h^6 :

$$\begin{aligned} \text{Term} &= \frac{1}{h^2} \left[c_1 \frac{h^6}{360} + c_2 \frac{64h^6}{360} \right] u^{(6)} = \frac{h^4}{360} \left[\frac{16}{12} + 64\left(-\frac{1}{12}\right) \right] u^{(6)} \\ &= \frac{h^4}{360} \left[\frac{16 - 64}{12} \right] u^{(6)} = \frac{h^4}{360} (-4) u^{(6)} = -\frac{1}{90} h^4 u^{(6)}. \end{aligned}$$

Bound: $\frac{M_6}{90} h^4$.

2. Rounding Error Bound: Sum of absolute coefficients in numerator: $|-1| + |16| + |-30| + |16| + |-1| = 64$. Denominator is $12h^2$.

$$|E_{\text{round}}| \leq \frac{64\epsilon}{12h^2} = \frac{16\epsilon}{3h^2}.$$

3. Minimizing the Total Error:

$$\psi(h) = \frac{M_6}{90} h^4 + \frac{16\epsilon}{3h^2}.$$

Set $\psi'(h) = 0$:

$$\begin{aligned} \frac{4M_6 h^3}{90} - \frac{32\epsilon}{3h^3} &= 0 \implies \frac{2M_6 h^6}{45} = \frac{32\epsilon}{3} \\ h^6 &= \frac{32 \cdot 45\epsilon}{6M_6} = \frac{240\epsilon}{M_6} \implies h_{\text{opt}} \approx \left(\frac{240\epsilon}{M_6} \right)^{1/6}. \end{aligned}$$

V.3 Comparison and Observation

- **Second-Order Case:** Optimal step size $h \sim \epsilon^{1/4}$, minimal error $E \sim \epsilon^{1/2}$.
- **Fourth-Order Case:** Optimal step size $h \sim \epsilon^{1/6}$, minimal error $E \sim \epsilon^{2/3}$.

Observation: Since machine epsilon ϵ is very small (typically 10^{-16}), $\epsilon^{2/3}$ is significantly smaller than $\epsilon^{1/2}$. This means the fourth-order method achieves much higher accuracy. Additionally, since $\epsilon^{1/6} > \epsilon^{1/4}$ for small ϵ , the optimal step size for the higher-order method is **larger**, meaning we can achieve better accuracy without taking an extremely small step size (which avoids catastrophic cancellation issues).

Note: As the final examinations approach, I am busy preparing for the exams, so the latex code is actually given by Gemini3, while the solution is all done by hand myself in advance.